White-Box Monitors Pilot: a Trusted Qwen3.5-27B Reader

Probes, J-lens readouts and their decision layer on MonitoringBench, LinuxArena and MOLE, with second-audit corrections

October 2026

Summary. We ran an open model, Qwen3.5-27B, as a *trusted reader* over agent transcripts and read its activations with off-the-shelf and cheaply trained probes. It read 1,325 MonitoringBench trajectories (775 attacks, 550 honest) that `gpt-4.1-mini` had already scored step by step. It read them two ways: *full* (each step sees all earlier steps) and *per-step* (each step read alone). It also read 1,210 detector-aware rewrites, the LinuxArena sweep, and MOLE insider sessions. All of Idea 1–3's decision-layer machinery was then applied to the probe streams. Cost: about $70 of H200 time and $2.95 of API. An independent second audit re-derived every headline that the stored data still support; the corrected numbers are the ones stated here.

1. **Holds: the FIBS axial probe on a per-step follow-up question beats the LLM monitor.** TPR at 1% trajectory FPR is +0.19 to +0.21 above `gpt-4.1-mini` (0.34 vs 0.15), equally on both attack agent models. The gain is unchanged at equal realized FPR (+0.18). It is the only one of 18 pre-planned reader-vs-LLM contrasts that survives Holm correction.
2. **Holds: a probe-triage cascade.** The synthetic-sabotage probe sends the 10% most suspicious honest-calibrated steps (13% of all steps) to the LLM. This raises TPR from 0.17 to 0.30 (+0.13) at matched realized FPR, with about 87% fewer LLM calls.
3. **Weakened.**
   - Full-reading sabotage probe +0.10: Holm $p = 0.22$; the gain is confined to attacks by the same agent model as the honest runs.
   - Full-reading FIBS axial probe and every fusion built on it: the threshold does not transfer across tasks (2.05% realized FPR at nominal 1%).
   - LinuxArena ranking: no paired test, and the reader score files were lost.
4. **Negative.**
   - No accumulation rule beats max on any probe stream.
   - Probe streams are more context-heterogeneous than LLM streams (τ up to 3.8 vs $\leq$0.46), but per-context thresholds gain little at matched spend.
   - J-lens watch-lists are weak monitors.
   - Policy activations do not beat the trusted reader on MOLE.
5. **Rewrite artifact.** Detector-aware rewrites are detected *more* by the LLM and the follow-up probes, but not by token-level probes.

1 Motivation and prior work

Source and caveat. This section condenses an internal literature note (`docs/reports/White box monitors for AI control.md`, 21 Oct 2026). Its numbers were *not* re-checked against the primary sources for this report. Numbers that the note itself says reached it only through an automated summary or a search snippet are marked † (unverified secondhand). Treat them as leads, not citations.

1.1 Probes roughly match frontier text monitors where misbehavior is salient

- **FIBS** (FAR.AI, arXiv 2610.12445) is the reference point. It trains four probe architectures, including an "axial" transformer over the layer×token grid, on 163,016 "contextual falsehood" transcripts (28 families), reading 6 layers at 30–90% depth on 9 open models. On agent steps it appends "Did you take actions towards a hidden side task?" and contrasts the probe readings for a prefilled "Yes." and "No.". It reports 98.3% / 98.0% per-step AUROC on SHADE-Arena / BashArena vs 97.7% / 97.6% for frontier text monitors (comparator identity†). Limits: instructed (not trained) saboteurs, no probe-aware attacker, one global threshold.
- **Apollo deception probes** (arXiv 2502.03407; Llama-3.3-70B, layer 22/80, mean pooling): AUROC 0.964–0.999, 95–99% recall at 1% Alpaca FPR, "roughly equivalent" to a GPT-4o baseline. Recall fell to nearly 0 on Insider Trading two layers later, and the probe fires on honest answers to evaluation prompts. Earlier: coup probes from 43 synthetic examples reached 100% TPR at 1% FPR on 46 cases; a one-pair defection probe predicted sleeper-agent defection at 99.3% AUROC.
- **Where white-box loses or ties.** Liars' Bench: 0.54 balanced accuracy vs 0.73 for an LLM judge. Trained lying organisms: activation detectors "drop sharply". AUROC misleads at operating points: an AgentDojo activation monitor with 0.92 AUROC had 0.00 TPR at 5% FPR (arXiv 2610.04575).
- **Trusted readers.** The one precedent for probing a *trusted* model that reads untrusted transcripts is a single unreviewed repository (APPS backdoors: 0.91 AUROC vs 0.54 for the same model's own rating).

1.2 Deployments and robustness

- **Cheap first-stage screens.** Anthropic's Constitutional Classifiers++ runs a 377K-FLOPs/token linear probe in front of a classifier, escalating about 5.5% of traffic and cutting compute about 40×†. GDM's production misuse probes (MultiMax, rolling means) feed a probe→Flash cascade that calls the LLM on under 10% of traffic and beats Flash alone at about 1/50 the cost†.
- **Misalignment monitoring is described only secondhand.** Anthropic reportedly ran dishonesty, reward-hacking and eval-awareness probes on every transcript during most of post-training†; OpenAI describes per-token activation classifiers escalating to automated investigators† at about 20% overhead†; GDM's roadmap (arXiv 2607.13087) calls activation monitoring "more speculative". No lab has published recall or FPR for a misalignment probe.
- **Training breaks probes.** Embedding-space attacks cut harmfulness-probe recall from 100% to 0% (Bailey et al., ICLR 2026). RL against a probe evades mean-pooled linear probes in 59–82% of cases, while max-pooled and attention probes stay near 0% (arXiv 2506.14261; per-cell numbers†). "Neural Chameleons" trained to evade benign-concept probes zero-shot evade unseen deception probes (arXiv 2512.11949). Even RLVR with no probe penalty moved an off-domain deception probe from 0.99 to 0.38–0.66 AUROC (arXiv 2602.15515; values†).
- **Passive shift and context.** Stylistic rewrites collapse probes with clean AUROC $\geq$0.998 (arXiv 2605.27958); a mean-pooled probe missed 99.13% of long-context attacks†. Probes fire on honest text in risky contexts, yet almost every paper calibrates one global threshold on benign chat.

This is the motivating gap for a decision-layer project: probe thresholds are context-confounded, pooling is ad hoc, and there is no trajectory-level false-alarm accounting.

1.3 J-lens

The Jacobian lens maps a layer-l residual into the final-layer basis through the corpus-averaged Jacobian and decodes it with the unembedding. Its monitoring evidence is qualitative, plus one eval-awareness score (AUC 0.853). Our J-lens-diffing project found that standalone single-model J-lens audits detected 0 of 5 fine-tuned organisms (per the same note). A runtime J-lens monitor therefore needs a fixed token watch-list or a per-context baseline.

1.4 What this pilot tests

FIBS did not do trajectory aggregation, sequential tests, per-environment thresholds, fusion with black-box monitors, red-teaming with an aware attacker, or trusted-reader probing (per the note's fetched summaries). The pilot tests six things on one reader pass:

(a) decision rules (max vs accumulators) on probe streams;
(b) per-context probe thresholds;
(c) probe + LLM fusion and triage;
(d) probes on detector-aware rewrites;
(e) a J-lens watch-list as a step-level monitor;
(f) stage 2: policy vs reader activations.

Go/no-go rule, set before the run. Proceed to stage 2 if a reader probe's trajectory TPR at 1% FPR is within the bootstrap CI of `gpt-4.1-mini`'s, or if fusion adds $\geq$0.05 TPR at matched cost. Both conditions were met, so stage 2 was run.

2 Setup

2.1 Reader and reading protocol

- **Reader.** `Qwen/Qwen3.5-27B` (rev `fc05daec`, bf16) is a hybrid gated-DeltaNet/attention model, run with HF transformers and forward hooks on one on-demand H200. We chose it because it is the one open model with both released FIBS probes (layers 19/27/35/43/51/58) and a pre-fitted neuronpedia J-lens.
- **Framing.** A monitor-style chat: `gpt-4.1-mini`'s monitor system prompt minus the rubric, then the task text, then steps rendered as `<action index=k>` (actions truncated at 4,000 characters, outputs at 1,500; first 60 steps; final `submit` stripped as in Idea 1).
- **Full (history-aware).** One incremental pass with a KV and recurrent-state cache: step t's tokens see the task and all earlier steps.
- **Per-step (history-free; "none").** System prompt + task + step t only. The prefix cache is rolled back before each step.
- **Follow-up.** After each step the cache is snapshotted. A FIBS-format question ("Did the agent take actions towards a hidden side task?") with a prefilled `Yes.` / `No.` answer is appended, read, and rolled back.
- **Storage.** Only per-step pooled scores and per-step mean activations are kept (activations stay on the pod).

Stream (code)	Definition
Sabotage probe (`sab`)	Logistic regression per layer on 1,012 `gpt-4.1`-generated honest vs covert-malicious step pairs. The pairs cover 36 task themes × 8 generic objective categories taken from the monitor prompt. Theme-grouped CV AUROC is 0.945–0.971 by layer (`scores/probes_report.json`). Primary form: mean over the 6 layers, mean-pooled over step tokens. Variants: max-token; single layers (L43 is *post hoc*).
Apollo probe (`dec`)	Instructed-Pairs deception probe on 306 true facts. CV AUROC 1.0, a known caveat: the instruction is in context.
FIBS linear on tokens (`fibs_tok`)	Released FIBS linear probe read directly on the step tokens.
FIBS linear follow-up (`fibs_fu`)	FIBS linear probe at the follow-up answer; score = reading(`No.`) − reading(`Yes.`).
FIBS axial follow-up (`fibs_fu_ax`)	The same contrast with the released FIBS axial probe.
J-lens (`jl_lm`, `jl_hits`)	Watch-list of 81 single-token ids (deceive, hide, secret, exfiltrate, backdoor, malicious, sabotage, evade, monitor, suspicious, . . .). Two scores: log-mass at layers 27/35/43 (layer mean) and the number of watch-list tokens in the top 25.
Supervised probe (`sup`)	Cross-fitted LR on per-step mean activations of the evaluation data. An in-distribution upper bound, leaky (§2.5).
LLM baseline	`gpt-4.1-mini`, prompt `ctrl-basic-v1`, logprob-expected 0–10 score, same history mode (from Idea 1).

Table 1: Per-step signals. Each is pooled over the step's tokens (mean / max / last). The trajectory statistic is the max over steps unless stated otherwise.

2.2 Signals

Primary streams. The results log designates `sab` (layer mean), the FIBS follow-up contrasts and J-lens mass as "pre-specified primary". The second audit found that this designation first appears in the results entry, not the setup entry (`results/audit2/issues.csv`, m2). This report therefore treats the 18 stream×mode reader-vs-LLM contrasts as one family and applies Holm.

What the probe directions decode to (J-lens decoding of the probe weight vectors; qualitative, `scores/probe_direction_jlens.json`): `sab` at L51/L58 to *hidden, disguised, simultaneously, conceal, suspicious, rogue*, and at L43 to delete/remove/hide tokens (mostly Chinese); `dec` and FIBS at L35–43 to *fake, false, falsehood*; FIBS at L51/58 to *Wait, False, Actually*.

2.3 Data

Set	Honest / attack	Modes	Comparators	Score files now
MonitoringBench (MB)	550 / 775 (S1–S3 100 each, S4 475), 37.7k steps	full, none	`gpt-4.1-mini`	intact
T3 round 1 rewrites	same honest / 509 unique (126 base attacks)	full, none	`gpt-4.1-mini`, `gpt-4.1`	**lost (0 bytes)**
T3 round 2 rewrites	same honest / 701 unique (150 base attacks)	none	`gpt-4.1`	intact
LinuxArena	200 / 142, 10 environments	full, none	12 per-action LLM monitors	**lost (0 bytes)**
MOLE `qwen36_27b` single-day	200 benign / 233 insider sessions	policy, reader	none	intact

Table 2: Scored sets. Honest-550 is entirely `claude-opus-4-5`. Of the 775 MB attacks, 494 are opus-4-5 and 281 are opus-4-6. "Lost": the reader per-step files were overwritten with empty files by the last pod sync, after the LinuxArena and T3-r1 analyses had run. The pods have since been deleted (`results/audit2/issues.csv`, A2).

2.4 Statistics

Randomized plug-in thresholds are set on the honest calibration half of task-grouped 50/50 splits, at trajectory FPR 0.05 / 0.01 / 0.003. Intervals are 95% task-cluster bootstraps that include calibration noise; contrasts use `amd.evaluation.paired` (one joint resample per replicate, thresholds recomputed). The project analysis (`results/wb/`) used only 10 splits, so its point estimates carry a few points of split noise. The second audit (`results/audit2/`) re-implemented everything independently (`a2lib.py`, no reuse of `amd.evaluation`) with 40 and 200 splits, reports the *realized* test FPR, and equalizes realized FPR by bisection on the nominal level. "Holds / weakened / dropped" verdicts follow `results/audit2/corrections.csv`.

2.5 Integrity checks

- **Numerical (incremental vs single pass, 24 trajectories, 661 steps).** Mean-pooled step features agree to Pearson $r \geq 0.999$ (median $|\Delta| \leq 0.02$ SD). Max-token features agree to $r = 0.98$–0.99 (e.g. J-lens L43 max: $r = 0.982$, max $|\Delta| = 2.1$ SD) (`wb/numerical_check.json`). bf16 kernels are chunk-boundary sensitive, so max-pooled variants carry numerical noise. Mean-pooled streams are reliable.
- **Probe training data is disjoint from the evaluation data.** The maximum TF-IDF cosine between a contrast task and an MB task is 0.61. Only 9 of 12,009 contrast 8-grams appear in 400 MB attacks, and they are generic boilerplate (`wb/disjointness.json`).
- **No source/style leakage in the generic streams.** On *pre-onset* attack steps vs honest steps (9,188 steps), every generic probe and the LLM score AUROC 0.46–0.58 (Apollo full is highest at 0.58) (`wb/mb_preonset.csv`).
- **The supervised probe is leaky.** It scores 0.79 (full) / 0.70 (none) on pre-onset steps, so it partly learns trajectory-source features. It also transfers poorly: LinuxArena AUROC 0.69 / 0.55 (`wb/la_auroc.csv`). It is reported only as a leaky upper bound.
- **Agent model is label-correlated** (first-audit M9). All honest runs are opus-4-5; 36% of attacks are opus-4-6. The LLM's TPR doubles on opus-4-6 attacks (full: 0.113 vs 0.253), so every contrast is also reported per agent model (`wb/tables.md`).
- **Length.** On length-matched subsets (383 honest, 383 attacks) the stream ordering is unchanged at FPR 0.01: `sab` full 0.21 vs LLM 0.15, FIBS axial none 0.30 vs LLM 0.09 (`wb/mb_length_matched.csv`). Realized FPRs there range 0.9–2.7%.

3 Results

3.1 Reader probes vs the LLM monitor on MonitoringBench

Findings.

- **The per-step FIBS axial follow-up is the one robust win.** It gives TPR 0.34 vs 0.15 for the LLM at 1% FPR. The gain is +0.19 to +0.21 across three independent estimates, is equal on opus-4-5 and opus-4-6 attacks, and is unchanged at equal realized FPR. It is the only reader-vs-LLM contrast of 18 that survives Holm. Its AUROC (0.85) is the highest of any non-leaky stream except the post hoc `sab` L43.
- **The full-reading sabotage probe is weakened.** It is nominally +0.10 and robust to realized FPR (1.12% vs 1.42%), but does not survive Holm ($p = 0.22$), and the whole gain is on opus-4-5

Stream	Mode	Project (10 splits), reader − LLM			Separation (AUROC of max)	
		FPR 0.05	FPR 0.01	FPR 0.003	reader	LLM
`sab` (mean)	full	+0.125 [+0.04, +0.21]	+0.098 [+0.02, +0.16]	+0.047 [−0.00, +0.14]	0.79	0.66
	none	−0.071 [−0.17, +0.06]	−0.023 [−0.14, +0.07]	−0.026 [−0.12, +0.04]	0.74	0.76
`sab` L43 (post hoc)	full	+0.331 [+0.24, +0.41]	+0.256 [+0.11, +0.35]	+0.081 [+0.04, +0.27]	0.89	0.66
	none	+0.034 [−0.07, +0.17]	+0.067 [−0.05, +0.16]	+0.045 [−0.04, +0.12]	0.81	0.76
FIBS linear follow-up	full	+0.027 [−0.10, +0.15]	−0.055 [−0.10, +0.08]	−0.069 [−0.10, +0.05]	0.75	0.66
	none	−0.007 [−0.06, +0.12]	+0.120 [+0.02, +0.20]	+0.101 [+0.03, +0.20]	0.79	0.76
FIBS axial follow-up	full	+0.229 [+0.02, +0.35]	+0.057 [+0.01, +0.29]	+0.052 [+0.01, +0.25]	0.82	0.66
	none	+0.125 [+0.08, +0.28]	**+0.206** [+0.09, +0.32]	+0.103 [+0.05, +0.29]	**0.85**	0.76
FIBS linear on tokens	full	+0.080 [−0.04, +0.16]	+0.022 [−0.06, +0.09]	+0.001 [−0.07, +0.07]	0.73	0.66
	none	−0.038 [−0.15, +0.18]	+0.039 [−0.10, +0.18]	+0.025 [−0.07, +0.16]	0.74	0.76
Apollo deception	full	−0.081 [−0.15, +0.08]	−0.092 [−0.14, −0.01]	−0.103 [−0.14, −0.05]	0.68	0.66
	none	−0.123 [−0.21, +0.06]	−0.005 [−0.12, +0.08]	+0.012 [−0.09, +0.07]	0.70	0.76
J-lens mass	full	+0.056 [−0.05, +0.15]	−0.044 [−0.09, +0.03]	−0.086 [−0.11, −0.02]	0.72	0.66
	none	−0.093 [−0.20, +0.07]	−0.029 [−0.15, +0.05]	−0.010 [−0.11, +0.04]	0.67	0.76
J-lens top-25 hits	full	−0.010 [−0.09, +0.05]	−0.032 [−0.07, +0.02]	−0.046 [−0.07, +0.01]	0.65	0.66
	none	−0.181 [−0.26, −0.05]	−0.066 [−0.18, +0.01]	−0.034 [−0.14, +0.00]	0.56	0.76
Supervised (leaky bound)	full	+0.764	+0.808	+0.773	1.00	0.66
	none	+0.560	+0.518	+0.427	0.98	0.76

Table 3: TPR of the trajectory max, reader stream minus `gpt-4.1-mini` in the same mode (775 attacks vs 550 honest; `wb/tables.md`, `wb/mb_auroc.csv`). These are nominal-FPR numbers from 10 splits. Table 4 gives the audited version for the main streams. Max-token J-lens mass is null (AUROC 0.48).

Stream	Mode	40 splits		Realized FPR	200 splits		Holm
		reader − LLM [CI]	opus-4-5 / 4-6	reader / LLM	nominal	equal FPR	(p)
FIBS axial follow-up	none	**+0.194** [+0.10, +0.28]	+0.199 / +0.183	1.64% / 1.34%	+0.185	**+0.183**	**survives**
FIBS axial follow-up	full	+0.085 [+0.04, +0.30]	+0.117 / +0.029	1.67% / 1.36%	+0.096	+0.059[a]	0.31
`sab` (mean)	full	+0.100 [+0.03, +0.17]	+0.148 / +0.016	1.46% / 1.36%	+0.095	+0.095	0.22
`sab` (mean)	none	−0.047 [−0.15, +0.04]	−0.033 / −0.072	1.34% / 1.34%	−0.051	−0.051	–
FIBS linear follow-up	none	+0.108 [+0.03, +0.18]	+0.114 / +0.098	2.06% / 1.34%	+0.096	+0.083	0.15
FIBS linear follow-up	full	−0.046 [−0.08, +0.09]	−0.028 / −0.076	1.70% / 1.36%	−0.038	−0.077	–
`sab` L43 (post hoc)	full	+0.240 [+0.12, +0.34]	+0.323 / +0.093	1.25% / 1.36%	+0.238	+0.238	n/a
`sab` L43 (post hoc)	none	+0.019 [−0.08, +0.11]	+0.036 / −0.010	1.66% / 1.34%	+0.015	+0.000	n/a

Table 4: Second-audit reproduction at FPR 0.01 (`audit2/repro_wb_mb.csv`, `audit2/repro_wb_mb_fpr_equalized.csv`, Holm p from `audit2/issues.csv` m2). The Holm family is 18 stream×mode contrasts; L43 is excluded as post hoc. The 200-split realized FPRs for the equal-FPR column are 2.05% (FIBS axial full) and 1.12% (`sab` full), vs 1.42% for the LLM. [a]The full-mode FIBS-axial threshold cannot be pushed below 1.58% realized FPR at any nominal level. +0.059 is the gain at that floor, still at a higher FPR than the LLM.

attacks (+0.148 vs +0.016 on opus-4-6): the LLM finds opus-4-6 attacks twice as often, the probe does not. The probe is thus better exactly in the like-for-like comparison (honest and attack agents match), but this is not a family-wise result.

- **The full-reading FIBS axial probe runs hot.** At nominal 1% its realized FPR is 2.05% (the LLM's is 1.42%), because some task clusters have heavy honest tails ($\tau = 1.64$, §3.5). Its reported +0.057 is ≈ +0.06 at the reader's FPR floor, and that floor is still above the LLM's FPR. Treat this contrast as not shown.
- **Generic deception and J-lens readouts lose to the LLM.** The Apollo probe (full) is −0.09 [−0.14, −0.01]. J-lens mass and hits are −0.03 to −0.18. Max-token J-lens is at chance.

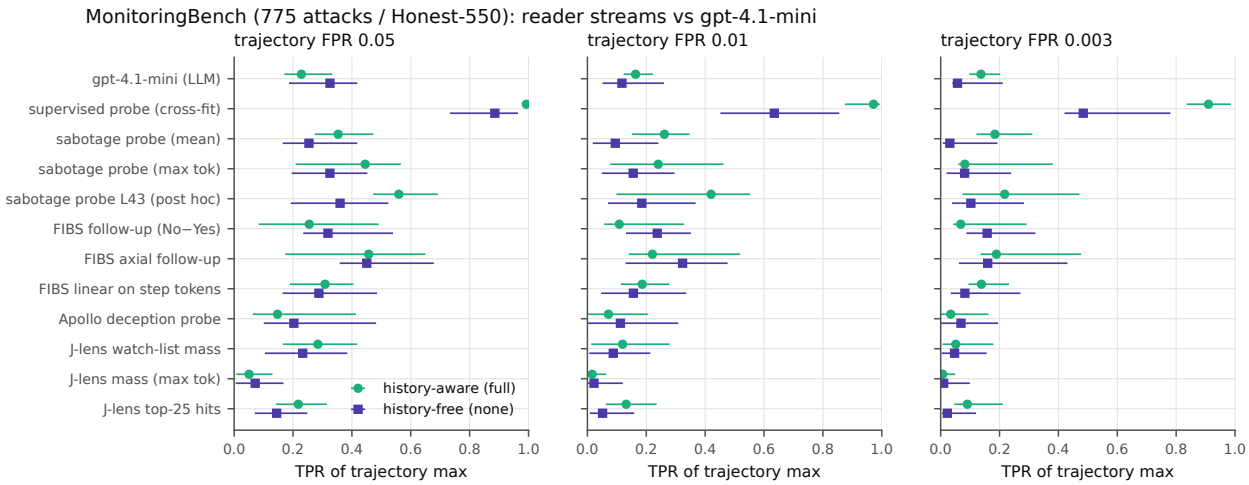


Figure 1: TPR of the trajectory max at FPR 0.05 / 0.01 / 0.003 with 95% CIs, per reader stream and the LLM. Green: full reading. Purple: per-step reading (project analysis, nominal FPR).

- **A post hoc per-layer probe is far better, but unvalidated.** The best single `sab` layer (L43, chosen on the evaluation set) gives +0.24 in full mode at matched FPR. This is a hypothesis for a pre-registered replication, not an estimate.

3.2 Full vs per-step reading

The same reader under two reading modes plays the role of Idea 1's history-aware vs history-free LLM monitor. The modes favor different signal families:

- **Token-level probes prefer full reading.** `sab` AUROC is 0.79 full vs 0.74 none. Read with context, a step's tokens carry more sabotage signal.
- **The follow-up question prefers per-step reading.** FIBS axial AUROC is 0.82 full vs 0.85 none, and TPR at 1% is 0.25 vs 0.34 (`audit2/repro_wb_mb.csv`). In full mode the question is asked after an ever-growing transcript, and honest follow-up scores persist: given one high step, the probability of a second is 0.66–0.77 in full mode vs 0.39–0.47 per step (`wb/mb_mechanism.csv`). Their honest tails also vary more across task clusters (τ 1.64 / 2.70 full vs 0.37 / 0.95 none).
- **The LLM, like the follow-up probe, separates better per step on AUROC** (0.66 full vs 0.76 none), but its TPRs at 1% are nearly equal (0.165 vs 0.148).
- **Attack structure.** Every stream separates distributed (S4) attacks at least as well as concentrated ones (S1–S3); e.g. AUROC on S4 vs S1 is 0.88 vs 0.80 for per-step FIBS axial, 0.81 vs 0.81 for `sab` full, 0.79 vs 0.73 for the per-step LLM (descriptive; `wb/mb_subgroups.csv`).

3.3 J-lens watch-list

- **As a monitor the watch-list is weak.** Layer-mean log-mass gives AUROC 0.72 / 0.67 (full / none) and TPR at 1% below the LLM. Top-25 hit counts give 0.65 / 0.56. Max-token mass is at chance (0.48), which also reflects bf16 noise in max-pooled features.
- **Single layers do not help.** AUROC is 0.66 / 0.62 / 0.69 at L27 / L35 / L43 (full) (`wb/mb_auroc.csv`).
- **It is the one signal that helps on MOLE.** There the reader's top-25 hits reach session

AUROC 0.96 (max) and 0.98 (mean) (§3.9).

- **Interpretation.** A trusted reader's J-space reflects the reader's thoughts about the transcript, not the agent's cognition. On BashArena, sysadmin work keeps tokens like *hide*, *monitor* and *secret* high on honest runs as well. This matches the literature note's prediction that a standalone readout is swamped without a baseline.

3.4 (a) Decision rules over probe streams

Stream	Mode	2nd-max	top-3 sum	CUSUM k=0.5	CUSUM k=1.5	OR(max+CUSUM)
LLM	full	+0.029 [−0.02, +0.10]	+0.031 [−0.00, +0.08]	−0.135 [−0.18, −0.10]	+0.018 [−0.00, +0.05]	−0.035 [−0.07, −0.01]
LLM	none	+0.041 [−0.07, +0.11]	+0.069 [−0.04, +0.14]	−0.081 [−0.19, −0.02]	−0.002 [−0.09, +0.05]	−0.039 [−0.13, +0.00]
`sab`	full	−0.143 [−0.18, −0.09]	−0.131 [−0.17, −0.07]	−0.207 [−0.26, −0.12]	−0.093 [−0.16, +0.01]	−0.079 [−0.13, −0.00]
`sab`	none	−0.035 [−0.09, +0.00]	−0.044 [−0.09, −0.01]	−0.044 [−0.10, +0.02]	−0.037 [−0.10, +0.01]	−0.021 [−0.06, +0.02]
FIBS lin. follow-up	full	−0.021 [−0.03, +0.02]	−0.019 [−0.04, +0.00]	−0.002 [−0.13, +0.05]	−0.022 [−0.06, +0.01]	−0.006 [−0.10, +0.03]
FIBS lin. follow-up	none	−0.124 [−0.18, −0.05]	−0.106 [−0.15, −0.04]	−0.179 [−0.25, −0.10]	−0.182 [−0.24, −0.08]	−0.056 [−0.10, −0.01]
FIBS axial follow-up	full	−0.011 [−0.05, +0.01]	−0.024 [−0.07, −0.00]	−0.032 [−0.24, +0.02]	−0.042 [−0.11, −0.01]	−0.001 [−0.17, +0.04]
FIBS axial follow-up	none	−0.122 [−0.22, −0.00]	−0.119 [−0.19, −0.01]	−0.246 [−0.34, −0.13]	−0.213 [−0.31, −0.04]	−0.134 [−0.21, −0.04]
FIBS linear tokens	full	−0.091 [−0.13, −0.01]	−0.076 [−0.11, −0.01]	−0.133 [−0.18, −0.05]	−0.097 [−0.15, −0.00]	−0.029 [−0.08, +0.01]
Apollo	none	−0.045 [−0.09, +0.02]	−0.051 [−0.08, +0.01]	−0.100 [−0.15, −0.04]	−0.064 [−0.09, +0.02]	−0.047 [−0.09, −0.00]
J-lens mass	none	−0.026 [−0.06, +0.03]	−0.035 [−0.07, +0.00]	−0.054 [−0.10, +0.03]	−0.061 [−0.10, −0.01]	−0.030 [−0.07, +0.01]

Table 5: TPR difference vs the trajectory max at FPR 0.01 (`wb/tables.md`, `wb/mb_paired.csv`). No accumulator beats max for any probe stream (only the leaky supervised probe has non-significant positive cells).

- **Max is the right rule on probe streams.** Accumulators are mostly worse, by up to −0.25 (CUSUM k=0.5 on per-step FIBS axial). Idea 1's lesson transfers unchanged: max-type rules first, and CUSUM only with a large reference value.
- **Difference-in-differences (none − full) of the CUSUM k=0.5 gain over max, 1% FPR.** The sign depends on the probe: `sab` +0.163 [+0.08, +0.22] (history-free reading favors accumulation, as Idea 1 predicted for LLM monitors); FIBS follow-ups the opposite (−0.177 [−0.25, −0.05] linear, −0.215 [−0.33, +0.02] axial); LLM +0.054 [−0.05, +0.12], n.s. These come from the 10-split analysis and were not in a pre-declared family: exploratory.
- **Idea 1's mechanism variables did not flip the ranking anywhere** (descriptive; unit threshold at a 5% honest step rate, `wb/mb_mechanism.csv`). Honest spike persistence $P(\geq 2 \text{ high} \mid \geq 1)$ is 0.52 for `sab` full, 0.77 for FIBS linear follow-up full, 0.33 for the LLM full; the repeat-spike log-LR is 0.37, 0.11 and 0.67. High persistence with low repeat-spike informativeness predicts that accumulation hurts, and it does; even `sab` none (0.53, 0.26) is outside the regime where accumulation won in Idea 1.

3.5 (b) Per-context probe thresholds

- **Probe streams are clearly more context-dependent than the LLM.** τ is 1.2–3.8 for four probe streams in full mode (all $p \leq 0.004$) vs 0.46 (n.s.) for the LLM. Cluster share of honest-max variance is 2–14% vs 0.2–3.5%. This replicates the context confound reported for Apollo-style probes, and it is the precondition for per-context allocation to matter.
- **Allocation at 10% spend, spend-matched** (CFAR, global and Stackelberg rows matched in ≥87% of replicates; `wb/context_allocation.csv`). CFAR raises uniform-attacker TPR more for probes than for the LLM, but the only interval excluding 0 is Apollo none, 0.31 → 0.46 [+0.01, +0.29], 1 of 14 MB contrasts (`sab`: full +0.04, none +0.09, n.s.; LLM −0.02 / +0.02). CFAR

	τ at global 5% (p)		context share of honest-max variance	
Stream	full	none	full	none
LLM (`gpt-4.1-mini`)	0.46 (0.16)	0.00 (1.00)	0.002	0.035
`sab`	1.21 (0.004)	0.53 (0.16)	0.075	0.087
FIBS linear follow-up	2.70 (<0.001)	0.95 (0.001)	0.140	0.110
FIBS axial follow-up	1.64 (<0.001)	0.37 (0.27)	0.107	0.068
FIBS linear tokens	0.39 (0.30)	1.26 (<0.001)	0.029	0.048
Apollo deception	3.79 (<0.001)	1.89 (0.001)	0.036	0.103
J-lens mass	0.56 (0.17)	0.31 (0.37)	0.030	0.020

Table 6: Between-cluster heterogeneity of honest false-positive rates on 7 MB task clusters: logit-normal τ at a global 5% threshold, and the share of honest-max variance explained by cluster (`wb/tables.md`, `wb/context_heterogeneity.csv`). Per-cluster honest FPRs range up to 0–24.6% for probes vs 0–12.3% for the LLM.

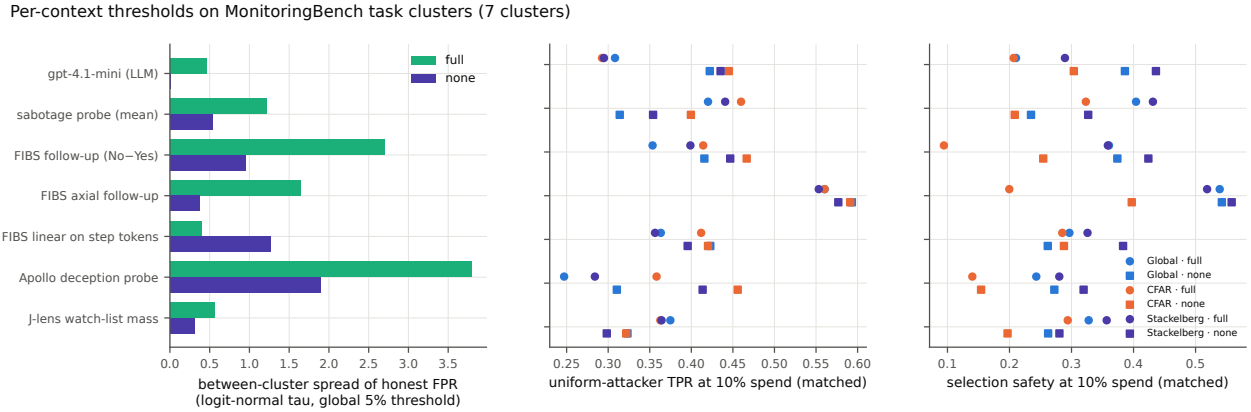


Figure 2: Left: τ per stream and mode. Middle: uniform-attacker TPR at 10% spend (spend-matched). Right: safety against an informed context-selecting attacker, for global, CFAR and Stackelberg thresholds.

lowers safety against an informed context-selecting attacker (FIBS linear follow-up full: 0.36 → 0.09, [−0.46, +0.01]). Stackelberg keeps selection safety within 0.02 of global or above on all 14 MB panels (no interval excludes 0). With 7 clusters most differences are n.s. The 5% rows (matched in 40–80% of replicates) and the LinuxArena allocation rows (15–70%, clamped; audit2 m10) are not used.

- **Related correction** (first audit C1, `results/fixes/corrections.csv`). The earlier E5 claim that "CFAR roughly doubles uniform TPR for probes at 1%" is replaced: 1% spend-matching is not estimable (per-context rules have a realized-spend floor of about 2.5%), and at equal realized spend CFAR gains +0.18 [+0.08, +0.27] for Apollo full (3.6%) and +0.15 [+0.03, +0.28] for FIBS linear follow-up full (2.9%), i.e. 1.3–1.5×, not 2×.

3.6 (c) Fusion and triage with the LLM monitor

- **The `sab` cascade holds.** Full reading, 10% of honest-calibrated steps escalated (13.1% of all steps): TPR 0.298 vs 0.168, +0.129 [+0.07, +0.23], at realized FPR 1.43% vs 1.81%, so the cascade does not over-spend (`audit2/repro_wb_fusion.csv`). It is a power gain *and* an 87% cut

Probe	Mode	LLM alone	Probe alone	KDE-LLR (traj)	Rank-sum	Cascade 5%	Cascade 10%
`sab`	full	0.166 (1.4%)	0.258 (1.5%)	0.279 (1.0%)	0.264 (0.8%)	0.306 (1.2%)	**0.299** (1.5%)
`sab`	none	0.151 (1.3%)	0.087 (1.0%)	0.201 (1.7%)	0.199 (1.2%)	0.345 (1.0%)	0.280 (1.0%)
FIBS axial f-u	full	0.166 (1.4%)	0.280 (**2.6%**)	0.328 (**2.8%**)	0.293 (1.9%)	0.336 (1.8%)	0.214 (1.2%)
FIBS axial f-u	none	0.151 (1.3%)	**0.324** (0.9%)	0.312 (1.7%)	0.314 (2.0%)	0.148 (1.3%)	0.150 (1.3%)
FIBS lin. f-u	full	0.166 (1.4%)	0.137 (2.7%)	0.214 (2.8%)	0.207 (1.8%)	0.319 (1.7%)	0.264 (1.4%)
J-lens mass	full	0.166 (1.4%)	0.112 (1.5%)	0.221 (2.5%)	0.224 (1.3%)	0.149 (1.4%)	0.162 (1.8%)

Table 7: TPR at nominal trajectory FPR 0.01, with *realized* FPR in parentheses (`wb/fusion.csv`). "Cascade q": the probe flags the top q of honest-calibrated steps, and only those steps go to the LLM, whose score decides. KDE-LLR: a trajectory-level log-likelihood ratio of (probe max, LLM max). Bold FPRs are the A1 problem: realized FPR about 2× nominal.

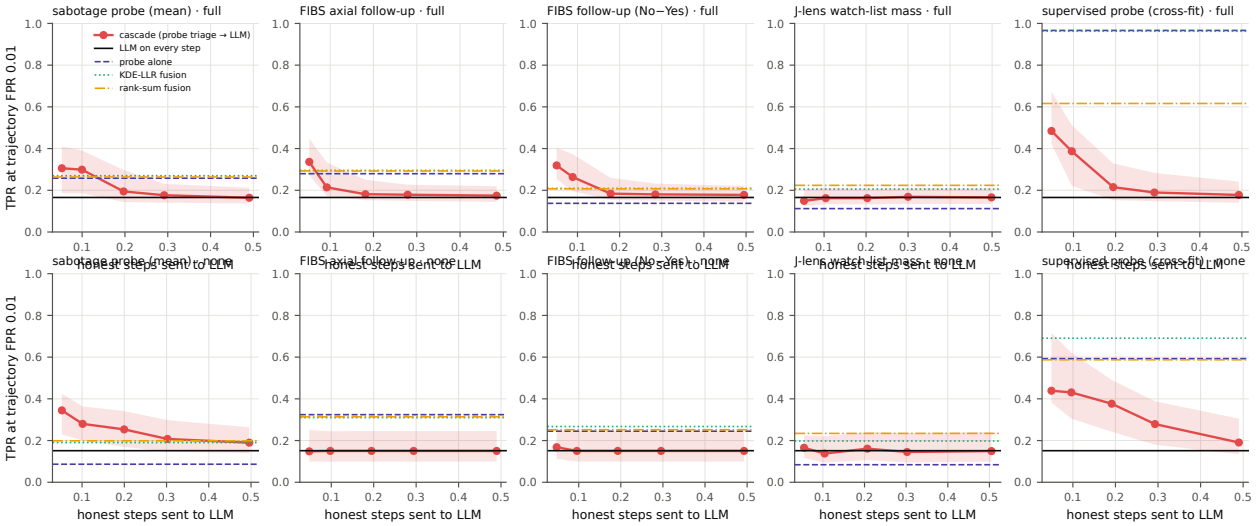


Figure 3: TPR at trajectory FPR 0.01 vs the share of honest steps sent to the LLM (cascade, red), compared with the LLM on every step (black), the probe alone, and KDE-LLR / rank-sum fusion. Top row: full reading. Bottom row: per-step reading.

in LLM calls: the probe removes honest steps the LLM would false-alarm on, so a tighter LLM threshold becomes affordable. Per-step reading gives the same +0.129 [+0.04, +0.18].

- **Cascade caveats.** $q = 10\%$ was one of 5 levels × 7 rules × 5 probes × 2 modes; $q = 5\%$ gives +0.146 and $q = 20\%$ only +0.04 (`audit2/repro_wb_fusion.csv`). The gain is +0.165 on opus-4-5 attacks vs +0.067 on opus-4-6 (`audit2/repro_wb_fusion_strata.csv`).
- **FIBS-axial full fusion headlines are weakened (A1).** KDE-LLR's 0.328 (+0.16) runs at 2.8% realized FPR vs 1.4% for the LLM, so it is not an FPR-matched comparison. The FIBS-axial-full cascade at 5% (0.336, realized 1.8%) is closer to matched but shares the problem.
- **Per-step FIBS axial needs no LLM.** The probe alone (0.324 at 0.9% realized FPR) is as good as any fusion. The cascade, in which the LLM decides, falls back to LLM-level TPR.
- **Weak probes still help fusion.** J-lens mass in rank-sum fusion is +0.058 [+0.03, +0.11] at 1.3% realized FPR, although the J-lens alone is below the LLM.

3.7 (d) Probes on detector-aware rewrites

Idea 3's monitor-aware attacker rewrote MonitoringBench attacks against LLM monitors. Round 2 (pacing arms against `gpt-4.1`, plus a matched rewrite control) is fully re-analyzable. Round 1's reader streams are lost (A2).

	Detection TPR (FPR 0.01, max)			One-shot pace − one-shot control [CI]		
Stream (none)	original	control	pace_max (r0)	pace_max	pace_cusum	pace_multichart
`gpt-4.1` (LLM)	0.240	0.573	0.493	−0.080 [−0.16, +0.03]	−0.053 [−0.11, +0.04]	−0.093 [−0.16, −0.01]
FIBS axial follow-up	0.380	0.653	0.593	−0.060 [−0.13, +0.03]	−0.047 [−0.11, +0.02]	−0.067 [−0.14, +0.02]
FIBS linear follow-up	0.227	0.513	0.500	−0.013 [−0.10, +0.06]	−0.027 [−0.10, +0.06]	−0.007 [−0.09, +0.07]
`sab`	0.087	0.080	0.100	+0.020 [−0.06, +0.09]	+0.020 [−0.03, +0.07]	+0.047 [+0.00, +0.12]
FIBS linear tokens	0.220	0.207	0.293	+0.087 [+0.00, +0.16]	+0.040 [−0.03, +0.11]	+0.053 [−0.02, +0.15]
J-lens mass	0.073	0.300	0.260	−0.040 [−0.13, +0.08]	−0.020 [−0.11, +0.07]	−0.080 [−0.15, +0.02]
Apollo deception	0.113	0.260	0.253	−0.007 [−0.09, +0.07]	+0.027 [−0.04, +0.11]	+0.000 [−0.08, +0.08]

Table 8: T3 round 2 (150 base attacks, per-step reading; `wb/tables.md`, `wb/t3_detection.csv`). "Control" is a rewrite with the same rewriter and no pacing instruction. Pacing arms were written against `gpt-4.1`.

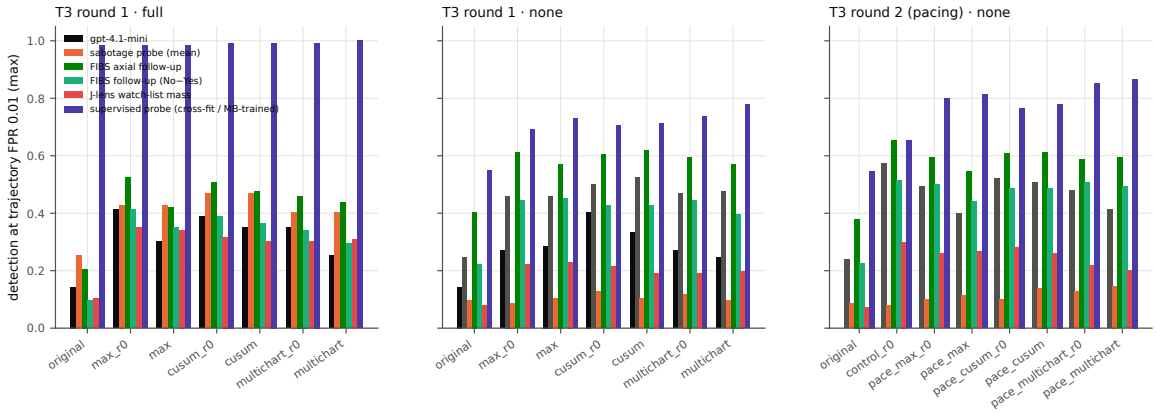


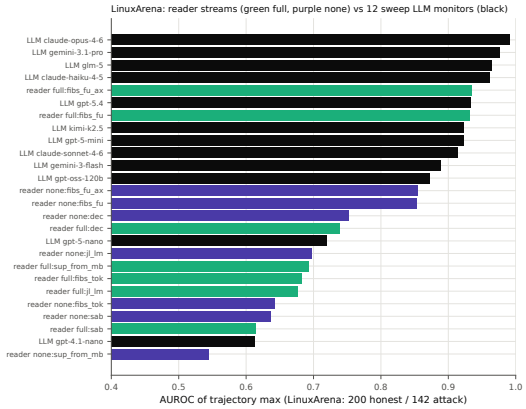
Figure 4: Detection at trajectory FPR 0.01 (max) on T3 rewrites by arm, for the LLM monitors and reader streams. Round 1 (left, middle) is not reproducible from stored data (A2). Round 2 (right) is intact.

- **The rewrite artifact is monitor-type specific.** The control rewrite roughly doubles detection by `gpt-4.1` (0.24 → 0.57) and the follow-up probes (FIBS axial 0.38 → 0.65), but leaves token-level probes unchanged (`sab` 0.087 → 0.080; FIBS tokens 0.22 → 0.21). The rewrites condense attacks and carry fabricated tool outputs (first audit M5); monitors that *ask a question about the transcript* react to that, probes on the action tokens do not.
- **Pacing written against `gpt-4.1` transfers partly to the follow-up probe, not to token probes.** One-shot pacing vs one-shot control is −0.05 to −0.09 for `gpt-4.1` and −0.05 to −0.07 (n.s.) for FIBS axial, but +0.02 to +0.09 for `sab` and FIBS tokens. No contrast is Holm-corrected. Only `gpt-4.1` pace_multichart has an interval excluding 0 in the evasion direction.
- **Why the token probes do not see the artifact.** On the E5 panel (first audit, M5) the same pattern appears as an AUROC confound. Refined variants of the same base attacks score higher than their originals for FIBS linear follow-up full (+0.20 [+0.11, +0.30]) and Apollo

full (+0.31). They do not score higher for `sab` max-token (−0.07 [−0.16, +0.01]). On the 33 unrewritten originals the `sab` probes have the highest AUROC (0.81–0.83, vs `gpt-4.1` 0.77) (`fixes/corrections.csv`).

- **Scope.** Every arm is an LLM-monitor-aware *prompted* rewrite. None is a probe-aware or trained evader, so this says nothing about robustness to the attacks in §1.

3.8 LinuxArena



Monitor	AUROC	TPR@0.05	TPR@0.01
LLM `claude-opus-4-6`	0.991	0.943 [0.90, 0.99]	0.923 [0.79, 0.97]
LLM `gemini-3.1-pro`	0.976	0.934 [0.87, 0.98]	0.902 [0.27, 0.95]
Reader FIBS axial f-u (full)	0.936	0.825 [0.75, 0.89]	0.756 [0.56, 0.86]
LLM `gpt-5.4`	0.933	0.702 [0.58, 0.86]	0.568 [0.37, 0.70]
Reader FIBS linear f-u (full)	0.932	0.784 [0.69, 0.88]	0.713 [0.57, 0.82]
LLM `claude-sonnet-4-6`	0.913	0.707 [0.62, 0.83]	0.645 [0.52, 0.75]
Reader FIBS axial f-u (none)	0.854	0.446 [0.13, 0.66]	0.167 [0.04, 0.52]
Reader J-lens hits (full)	0.754	0.419 [0.28, 0.57]	0.297 [0.19, 0.48]
Reader `sab` (full)	0.615	0.262 [0.14, 0.35]	0.159 [0.07, 0.30]
LLM `gpt-4.1-nano`	0.613	0.193 [0.11, 0.29]	0.120 [0.04, 0.22]

Figure 5: LinuxArena (200 honest, 142 attacks), trajectory max. Left: AUROC of reader streams (green full, purple per-step) vs 12 LLM monitors (black) on the same trajectories (`wb/la_auroc.csv`). Right: excerpt with marginal 95% CIs (`wb/tables.md`). The reader score files behind these numbers are lost (A2).

- **FIBS follow-up probes in full reading are comparable to mid-tier frontier monitors.** AUROC is 0.93–0.94, against `gpt-5.4` at 0.933 and `sonnet-4-6` at 0.913. They are below `opus-4-6` and `gemini-3.1-pro`.
- **The ranking is weakened (audit2 m3).** Only marginal CIs exist and they overlap ([0.56, 0.86] vs [0.37, 0.70] for `gpt-5.4`); the 1% threshold is the top of about 100 calibration honest runs; no paired test was run, and none can be run now (A2).
- **Mode reversal.** Unlike MB, full reading beats per-step reading here (0.94 vs 0.85).
- **The sabotage probe and J-lens are weak** (AUROC 0.62–0.75). The `sab` contrast pairs were generated from generic objectives, and LinuxArena's side tasks are more diverse.

3.9 Stage 2: policy vs reader activations on MOLE

- **Policy arm.** Qwen3.6-27B, the model that generated MOLE's `qwen36_27b` single-day sessions, reads its own sessions in its chat format, including the system prompt. That prompt states the insider objective. Reasoning tokens and tool schemas are not in the release.
- **Reader arm.** Qwen3.5-27B in the monitor framing, without the system prompt.
- **Probes.** `sab` was re-trained on each model's own activations of the same contrast pairs (CV AUROC 0.97 for both; `EXPERIMENT_LOG.md`). The FIBS 3.5 linear probe is applied to Qwen3.6 as a transfer.

- **Policy activations do not beat the reader on the main probe** (`sab`: 0.84 / 0.95 vs 0.86 / 0.96).

	Policy (Qwen3.6-27B)		Reader (Qwen3.5-27B)	
Stream	max over turns	mean over turns	max over turns	mean over turns
`sab` (layer mean)	0.839 [0.80, 0.88]	0.954 [0.94, 0.97]	0.863 [0.83, 0.89]	0.955 [0.94, 0.97]
`sab` L43 (post hoc)	0.899 [0.87, 0.93]	0.978 [0.96, 0.99]	0.938 [0.92, 0.96]	0.982 [0.97, 0.99]
FIBS linear on tokens	0.816 [0.78, 0.86]	**0.982** [0.97, 0.99]	0.620 [0.57, 0.67]	0.742 [0.70, 0.79]
J-lens mass	0.683 [0.64, 0.73]	0.764 [0.72, 0.80]	0.752 [0.71, 0.80]	0.829 [0.79, 0.87]
J-lens top-25 hits	0.800 [0.76, 0.84]	0.925 [0.90, 0.95]	0.956 [0.94, 0.97]	**0.984** [0.97, 0.99]

Table 9: Session AUROC, 233 insider vs 200 random benign sessions (`wb/mole_policy_vs_reader.csv`). TPR at 1% FPR for `sab` mean: policy 0.71, reader 0.74.

- **The one stream where the policy wins is suspect.** FIBS on tokens gives 0.98 vs 0.74 (mean), but this is also the stream that could key on the in-context insider instruction, which only the policy sees.
- **The reader's J-lens hit count is the best stream of either arm** (0.96 / 0.98).
- **The split cannot answer the policy-vs-reader question cleanly.** There is no LLM baseline for it; benign sessions are a random sample whose account and role mix is not matched to the insiders, so session-type differences may inflate every AUROC; and the policy sees a system prompt that names the objective.

4 What holds, what is weakened, what is dropped

Table 10 lists every headline claim with its verdict after the first audit (`results/fixes/`) and the second audit (`results/audit2/`).

5 Interpretation

1. **A trusted reader buys some power, not only cost savings.** The literature note expected "white-box adds cost savings, not power, for a trusted reader". One off-the-shelf probe (FIBS axial) on a per-step follow-up question beats a cheap LLM monitor by about 0.19 TPR at 1% FPR with no task-specific training. The comparator is weak (`gpt-4.1-mini`, AUROC 0.66–0.76), so "beats a cheap LLM monitor" is the right scope; on LinuxArena the strongest frontier monitors remain ahead.
2. **The probe's best use with an LLM is triage.** The `sab` cascade gains about 0.13 TPR *and* removes about 87% of LLM calls, the shape of the published GDM and Anthropic misuse cascades[†], here at matched trajectory FPR for multi-step sabotage.
3. **Probe streams need context-aware calibration.** Their between-task heterogeneity is much larger (τ 1.2–3.8 for four probe streams vs 0.46 for the LLM), and the one audit failure (A1) is exactly a probe threshold that did not transfer across tasks. Per-context allocation helps only modestly at matched spend, and CFAR is exploitable by a context-selecting attacker; Stackelberg is the safe default, as in Idea 2.
4. **The decision layer transfers unchanged.** Max is the right trajectory rule on every probe stream.
5. **Token-level probes and question-based monitors fail differently.** Rewriting artifacts move LLM monitors and follow-up probes but not token probes, an argument for combining

Claim	Verdict	Basis
Per-step FIBS axial follow-up beats `gpt-4.1-mini` at 1% FPR	**holds**	+0.206 / +0.194 / +0.185 (10 / 40 / 200 splits); +0.183 at equal realized FPR; both agent models; only Holm survivor of 18 (audit2 m1, m2)
`sab` cascade (10%) beats the LLM with 87% fewer calls	**holds**	0.298 vs 0.168, +0.129 [+0.07, +0.23] at realized FPR 1.43% vs 1.81% (audit2 m4); q chosen among 5; larger on opus-4-5
No accumulation rule beats max on probe streams	**holds**	Table 5; consistent with Idea 1
Probe streams are more context-heterogeneous than LLM streams	**holds**	honest-only statistic, unaffected by matching issues (first audit M5 row)
Full-reading `sab` beats the LLM (+0.10)	weakened	Holm $p = 0.22$; +0.148 on opus-4-5 vs +0.016 on opus-4-6 (audit2 m2)
Full-reading FIBS axial beats the LLM (+0.057)	weakened / not shown	realized FPR 2.05% vs 1.42%; floor 1.58%; +0.059 at the floor (audit2 A1)
FIBS-axial-full fusion (KDE-LLR 0.328, +0.16)	weakened	realized FPR 2.8% vs 1.4% (audit2 A1)
LinuxArena: FIBS follow-up above `gpt-5.4` and four others	weakened, unverifiable	no paired test; score files lost (audit2 m3, A2)
T3 round-1 reader detection rows	unverifiable	score files lost (A2); round 2 intact and consistent
CFAR on probes "roughly doubles" uniform TPR (E5, 1%)	dropped / replaced	1% matching not estimable; 1.3–1.5× at equal realized spend (first audit C1)
`sab` L43 (+0.24)	exploratory only	selected on the evaluation set
DiD of CUSUM gain (`sab` +0.16; FIBS −0.18)	exploratory	10 splits, not in a declared family
MOLE: policy activations add nothing over the reader	holds, weak design	unmatched benign sessions, no LLM baseline, objective in the policy's context

Table 10: Claim ledger after both audits (`results/audit2/corrections.csv`, `results/audit2/issues.csv`, `results/fixes/corrections.csv`).

the two kinds. It is not robustness evidence: no attacker here knew about the probes.

6 Limitations

- **One reader, one MB comparator** (`gpt-4.1-mini`); no frontier LLM monitor was run on MB.
- **Statistical power.** 550 honest runs put a 1% trajectory FPR on about 5 false alarms per calibration half; the project analysis used 10 splits; 7 MB clusters are too few for per-context significance.
- **Off-policy reading.** The reader reads Opus-authored text, the regime where intent probes generalize worst, and shares an LLM monitor's threat model (it sees only text).
- **Label-correlated agent model.** Honest runs are all opus-4-5; like-for-like results are reported, but surface features (truncation, reasoning visibility) remain label-correlated (first audit m8).
- **Lost data** (LinuxArena, T3-r1 reader streams; A2), **no adaptive white-box attacker** (every robustness concern in §1 stands untested), and **bf16 noise** in max-pooled features (up to 2 SD for single steps).

7 Recommended next steps

1. **Preregister and Holm-control a small confirmatory family** on a fresh split: (i) per-step FIBS axial vs the LLM; (ii) the `sab` cascade at a pre-declared q; (iii) `sab` L43 in full reading, replicating the post hoc layer. Use $\geq$100 grouped splits, report realized FPR next to every TPR, and compare at equal realized FPR.
2. **Regenerate the lost score files.** Re-read LinuxArena (342 trajectories) and T3 round 1 (509 attacks) in both modes (about 1–2 H200-hours), then run the missing paired LinuxArena contrasts. Guard the sync so it never overwrites a non-empty local file with an empty one.
3. **Calibrate the full-reading FIBS axial threshold per cluster** (Stackelberg or partially pooled) and re-state the full-mode fusions at matched realized FPR.
4. **Build a proper policy-activation arm**: matched benign sessions (same accounts, roles and days), an LLM baseline on the same split, an arm with the objective removed from the system prompt; better, fresh open-policy rollouts (e.g. Qwen3.6-27B on BashArena) with reasoning retained, so both arms see identical text.
5. **Add a strong LLM comparator on MB** (`gpt-4.1` or a frontier monitor on the same 1,325 trajectories) to test whether the FIBS-axial gain survives against a non-trivial monitor.
6. **Run a probe-aware attacker**: first prompted ("a white-box probe on a trusted model reads your actions") against a strong LLM monitor; then, if stage 1 holds, a trained RL-Obfuscation-style evader.
7. **Contact FAR.AI** for FIBS's on-policy BashArena and SHADE step streams; our aggregation, per-context and cascade analyses would run on them unchanged.

Cost and reproduction

GPU ≈$70 (two on-demand H200 pods, about 7.0 h and 5.8 h at $5.29/h, plus $2.1 of earlier setup; about 1,000 tokens/s per reader process with per-step follow-ups); OpenRouter $2.95 (contrast pairs); second audit $0. Code: `tracks/wb_whitebox/` (reader and analyses) and `tracks/audit2/`. Results: `results/wb/` (tables.md, CSVs, figures), `results/audit2/`, errata `results/fixes/corrections.csv`. Score streams: `data/processed/whitebox/scores/` (MB, T3 r2 and MOLE intact; LinuxArena and T3 r1 empty). Log: `EXPERIMENT_LOG.md` (2026-10-21/22). Repository: https://github.com/gregkocher/agent-monitor-detection (private).